Data dictionary — EnergyMonitor

Project: Electrical Consumption Monitor (EnergyMonitor) · SENA ADSO, group 3145556 **Schema:** `energy_monitor` (MySQL 8.4, InnoDB, utf8mb4) · **Updated:** October 27, 2026

1. Scope and source

This dictionary describes **every table and column** of the system database as it stands after applying all the Liquibase changesets of the seven services that have a database:

Service (port)	Repository	Changesets	Tables
security (8082)	en-monitor-security-service	security-001 to security-030	16
device (8084)	en-monitor-device-service	device-001 to device-009	5
measurement (8085)	en-monitor-measurement-service	measurement-001 to measurement-004	3
home (8086)	en-monitor-home-service	home-001 to home-007	5
notification (8087)	en-monitor-notification-service	notification-001 to notification-007	4
alert (8088)	en-monitor-alert-service	alert-001 to alert-004	2
recommendation (8089)	en-monitor-recommendation-service	recommendation-001 to recommendation-002	2
Total		**63 changesets**	**37**

The source of truth is the `src/main/resources/db/changelog/**` files of each service; this document and the script `../../script-DB/energy_monitor.sql` were generated from them. The API gateway (8080) has no database. The retired `energy-monitor-backend` no longer creates tables: the ones it owned moved, with the same changesets and the same path, to the home, notification, alert and recommendation services, and a migrated database may still hold its `backend_outbox` table, unused. The Liquibase control tables (`DATABASECHANGELOG` and `DATABASECHANGELOGLOCK`) are not described.

2. Model rules

- **One shared schema, one owner per table.** Each table is created, migrated and written by a single service. The others get that data through the internal API (`/internal/**` with the `X-Internal-Token` header) or through RabbitMQ events.
- **Foreign keys only inside a service.** References to tables of another service (`user_id`, `home_id`, `device_id`, `measurement_id`, `consumption_level_id`) are `VARCHAR(10)` columns without a `FOREIGN KEY`, validated by the application; the *Constraint* column shows them as "application reference, no FK".
- **Identifiers** `VARCHAR(10)` generated by the application (for example `user000001`).
- **Audit and soft delete:** business tables have `created_at`, `updated_at` and `deleted_at`; a row with `deleted_at` counts as deleted.
- **Transactional outbox:** every service that publishes events has its `*_outbox` table (security, device, measurement, home, alert and recommendation). The notification service only consumes events and has no outbox.
- **Dates:** `TIMESTAMP` in UTC; the newer tables (outboxes and `pending_registration`) use `DATETIME(6)` with microseconds.

3. Table summary

#	Table	Owner service	Description	Primary key	Foreign keys (inside the service)
1	person	security (8082)	Records the personal data associated with a user.	id_person	—
2	user	security (8082)	Manages the user's credentials, status and access data.	id_user	person_id → person
3	user_configuration	security (8082)	Stores each user's configuration preferences.	id_configuration	user_id → user
4	password_reset_token	security (8082)	Stores tokens for password recovery.	id_reset_token	user_id → user
5	user_session	security (8082)	Records the user's active and renewable sessions.	id_user_session	user_id → user
6	security_configuration	security (8082)	Defines configurable security parameters of the system.	id_security_configuration	—
7	password_policy	security (8082)	Defines the password validation and expiration rules.	id_password_policy	—
8	login_error_log	security (8082)	Records the errors produced during login.	id_login_error	user_id → user
9	system_role	security (8082)	Defines the roles available in the system.	id_system_role	—
10	permission	security (8082)	Defines the functional permissions of the system.	id_permission	—
11	user_system_role	security (8082)	Relates users to system roles.	user_id, system_role_id	user_id → user; system_role_id → system_role
12	system_role_permission	security (8082)	Relates roles to permissions.	system_role_id, permission_id	system_role_id → system_role; permission_id → permission
13	audit_log	security (8082)	Records the relevant actions executed by users.	id_audit_log	user_id → user
14	refresh_token	security (8082)	Each row is a refresh token generation.	id_refresh_token	id_user_session → user_session; parent_id → refresh_token
15	security_outbox	security (8082)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—
16	pending_registration	security (8082)	Registrations waiting for the email verification.	id_user	—
17	appliance_type	device (8084)	Catalog of appliance types.	id_appliance_type	—
18	device	device (8084)	Records the installed measurement devices.	id_device	appliance_type_id → appliance_type
19	device_status_log	device (8084)	Records the status and connectivity of the devices.	id_device_status	device_id → device
20	device_home	device (8084)	Relates devices to households.	device_id, home_id	device_id → device

#	Table	Owner service	Description	Primary key	Foreign keys (inside the service)
21	device_outbox	device (8084)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—
22	measurement	measurement (8085)	Stores the recorded electrical measurements.	id_measurement	—
23	consumption_level	measurement (8085)	Defines risk levels and their consumption ranges.	id_consumption_level	—
24	measurement_outbox	measurement (8085)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—
25	home_type	home (8086)	Catalog of household types.	id_home_type	—
26	home	home (8086)	Records the households managed by the system.	id_home	home_type_id → home_type
27	home_thresholds	home (8086)	Configures a household's consumption limits.	id_threshold	home_id → home
28	user_home	home (8086)	Relates users to households and defines their role.	user_id, home_id	home_id → home
29	home_outbox	home (8086)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—
30	notification	notification (8087)	Record of the delivery of a message to the user.	id_notification	—
31	notification_preference	notification (8087)	Notification channel preferences of each user (email and push).	user_id	—
32	push_subscription	notification (8087)	Push notification subscriptions of the user's browsers.	id_push_subscription	—
33	push_device	notification (8087)	Mobile app installations that receive push notifications through the Expo relay (one row per installation).	id_push_device	—
34	alert	alert (8088)	Alerts of a home: reading above the power threshold (THRESHOLD, level HIGH or CRITICAL), daily or monthly limit reached (LIMIT), device offline (CONNECTIVITY) and first reading of a newly linked device (DEVICE).	id_alert	—
35	alert_outbox	alert (8088)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—
36	recommendation	recommendation (8089)	Saving recommendations the recommendation service generates every hour from the hourly energy of the home; device_id empty when it applies to the whole home.	id_recommendation	—
37	recommendation_outbox	recommendation (8089)	Transactional outbox of the service: events it publishes to RabbitMQ.	id_message	—

4. Dictionary by table

Legend: PK = primary key; FK = foreign key inside the same service; UNIQUE = uniqueness constraint (with its name); DEFAULT = default value; *Null* = Yes when the column accepts NULL. For ENUMs the *Size / values* column lists the allowed values; in `TIMESTAMP(6)` and `DATETIME(6)`, 6 is the microsecond precision.

person

Records the personal data associated with a user.

Owner service: en-monitor-security-service · **Changesets:** security-001-create-person, security-018-drop-person-contact-data, security-020-drop-person-cellphone

Field	Type	Size / values	Null	Constraint	Description
id_person	VARCHAR	10	No	PK	Person identifier
name	VARCHAR	100	No	—	Person's first names
last_name	VARCHAR	100	Yes	—	Person's last name (optional)
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

user

Manages the user's credentials, status and access data.

Owner service: en-monitor-security-service · **Changesets:** security-002-create-user, security-017-add-user-profile-image, security-022-widen-user-profile-image, security-024-add-user-google-subject, security-029-add-user-terms-accepted-at

Field	Type	Size / values	Null	Constraint	Description
id_user	VARCHAR	10	No	PK	User identifier
person_id	VARCHAR	10	No	UNIQUE (uk_user_person_id); FK → person.id_person	Associated person
password_hash	VARCHAR	255	No	—	Password hash
email	VARCHAR	255	No	UNIQUE (uk_user_email)	Email address
email_verified	BOOLEAN	—	No	—	Indicates whether the email was verified
registration_date	TIMESTAMP	—	No	—	Registration date and time
status	ENUM	ACTIVE, INACTIVE, BLOCKED	No	—	Account status
failed_login_attempts	INT	—	No	—	Number of failed attempts
last_login_at	TIMESTAMP	—	Yes	—	Last login
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)

Field	Type	Size / values	Null	Constraint	Description
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current
profile_image	MEDIUMTEXT	—	Yes	—	Account profile image (base64 data URL)
google_subject	VARCHAR	255	Yes	UNIQUE (uq_user_google_subject)	Stable identifier (sub) of the Google account the account was created with; empty if registered with a password
terms_accepted_at	TIMESTAMP	—	Yes	—	When the account accepted the terms and conditions; empty forces the acceptance at the next sign-in (answer 428)

Indexes: None

user_configuration

Stores each user's configuration preferences.

Owner service: en-monitor-security-service · **Changesets:** security-003-create-user-configuration

Field	Type	Size / values	Null	Constraint	Description
id_configuration	VARCHAR	10	No	PK	Configuration identifier
user_id	VARCHAR	10	No	UNIQUE (uk_user_configuration_user_id); FK → user.id_user	Configured user
notify_by_email	BOOLEAN	—	No	—	Enables email notifications
notify_by_push	BOOLEAN	—	No	—	Enables push notifications
color_theme	VARCHAR	50	Yes	—	Color theme
language	VARCHAR	50	Yes	—	Preferred language
social_provider	VARCHAR	50	Yes	—	Social authentication provider
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

password_reset_token

Stores tokens for password recovery.

Owner service: en-monitor-security-service · **Changesets:** security-004-create-password-reset-token, security-021-add-password-reset-token-attempts

Field	Type	Size / values	Null	Constraint	Description
id_reset_token	VARCHAR	10	No	PK	Token identifier

Field	Type	Size / values	Null	Constraint	Description
user_id	VARCHAR	10	No	FK → user.id_user	Owner user
reset_token_hash	VARCHAR	64	No	UNIQUE (uk_password_reset_token_reset_token_hash)	HMAC-SHA256 hash (with pepper) of the recovery code
used	BOOLEAN	—	No	—	Indicates whether the token was used
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
expiration_at	TIMESTAMP	—	No	—	Expiration date
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current
attempts	INT	—	No	DEFAULT 0	Attempts to redeem the code

Indexes: None

user_session

Records the user's active and renewable sessions.

Owner service: en-monitor-security-service · **Changesets:** security-005-create-user-session, security-014-add-session-revocation, security-016-drop-session-refresh-token

Field	Type	Size / values	Null	Constraint	Description
id_user_session	VARCHAR	10	No	PK	Session identifier
user_id	VARCHAR	10	No	FK → user.id_user	Session user
revoked	BOOLEAN	—	No	—	Indicates whether the session was revoked
ip_address	VARCHAR	45	Yes	—	IP address
user_agent	VARCHAR	150	Yes	—	Client agent
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
expiration_at	TIMESTAMP	—	No	—	Expiration date
closed_at	TIMESTAMP	—	Yes	—	Closing date
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current
revoked_at	TIMESTAMP	—	Yes	—	Date and time of session revocation

Field	Type	Size / values	Null	Constraint	Description
revoked_reason	ENUM	REFRESH_TOKEN_REUSE, PASSWORD_CHANGED, ADMINISTRATIVE	Yes	—	Reason for revocation

Indexes: None

security_configuration

Defines configurable security parameters of the system.

Owner service: en-monitor-security-service · **Changesets:** security-006-create-security-configuration

Field	Type	Size / values	Null	Constraint	Description
id_security_configuration	VARCHAR	10	No	PK	Configuration identifier
config_name	VARCHAR	100	No	UNIQUE (uk_security_configuration_config_name)	Configuration name
config_value	VARCHAR	100	No	—	Configuration value
description	VARCHAR	255	Yes	—	Description
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

password_policy

Defines the password validation and expiration rules.

Owner service: en-monitor-security-service · **Changesets:** security-007-create-password-policy

Field	Type	Size / values	Null	Constraint	Description
id_password_policy	VARCHAR	10	No	PK	Policy identifier
min_length	INT	—	No	—	Minimum length
max_length	INT	—	No	—	Maximum length
require_uppercase	BOOLEAN	—	No	—	Requires uppercase letters
require_numbers	BOOLEAN	—	No	—	Requires numbers
require_symbols	BOOLEAN	—	No	—	Requires symbols
expiration_days	INT	—	No	—	Days of validity
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)

Field	Type	Size / values	Null	Constraint	Description
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

login_error_log

Records the errors produced during login.

Owner service: en-monitor-security-service · **Changesets:** security-008-create-login-error-log, security-027-login-error-email-not-verified

Field	Type	Size / values	Null	Constraint	Description
id_login_error	VARCHAR	10	No	PK	Error identifier
user_id	VARCHAR	10	Yes	FK → user.id_user	Related user, if any
error_type	ENUM	INVALID_PASSWORD, USER_NOT_FOUND, ACCOUNT_BLOCKED, ACCOUNT_INACTIVE, EMAIL_NOT_VERIFIED	No	—	Error type
description	VARCHAR	500	Yes	—	Error description
ip_address	VARCHAR	45	Yes	—	IP address
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

system_role

Defines the roles available in the system.

Owner service: en-monitor-security-service · **Changesets:** security-009-create-system-role

Field	Type	Size / values	Null	Constraint	Description
id_system_role	VARCHAR	10	No	PK	Role identifier
name	VARCHAR	50	No	UNIQUE (uk_system_role_name)	Role name
description	VARCHAR	200	Yes	—	Description

Field	Type	Size / values	Null	Constraint	Description
enabled	BOOLEAN	—	No	—	Indicates whether it is enabled
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

permission

Defines the functional permissions of the system.

Owner service: en-monitor-security-service · **Changesets:** security-010-create-permission

Field	Type	Size / values	Null	Constraint	Description
id_permission	VARCHAR	10	No	PK	Permission identifier
code	VARCHAR	30	No	UNIQUE (uk_permission_code)	Unique code
name	VARCHAR	50	No	—	Permission name
description	VARCHAR	100	Yes	—	Description
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

user_system_role

Relates users to system roles.

Owner service: en-monitor-security-service · **Changesets:** security-011-create-user-system-role

Field	Type	Size / values	Null	Constraint	Description
user_id	VARCHAR	10	No	PK; FK → user.id_user	Related user
system_role_id	VARCHAR	10	No	PK; FK → system_role.id_system_role	Assigned role
assigned_at	TIMESTAMP	—	No	—	Assignment date
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)

Field	Type	Size / values	Null	Constraint	Description
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

system_role_permission

Relates roles to permissions.

Owner service: en-monitor-security-service · **Changesets:** security-012-create-system-role-permission

Field	Type	Size / values	Null	Constraint	Description
system_role_id	VARCHAR	10	No	PK; FK → system_role.id_system_role	Related role
permission_id	VARCHAR	10	No	PK; FK → permission.id_permission	Assigned permission
assigned_at	TIMESTAMP	—	No	—	Assignment date
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

audit_log

Records the relevant actions executed by users.

Owner service: en-monitor-security-service · **Changesets:** security-013-create-audit-log

Field	Type	Size / values	Null	Constraint	Description
id_audit_log	VARCHAR	10	No	PK	Record identifier
user_id	VARCHAR	10	Yes	FK → user.id_user	User who executed the action
action	ENUM	CREATE, UPDATE, DELETE, LOGIN, LOGIN_FAILED, LOGOUT	No	—	Audited action
description	VARCHAR	255	Yes	—	Description
ip_address	VARCHAR	45	Yes	—	IP address
application	VARCHAR	255	Yes	—	Source application
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

refresh_token

Each row is a refresh token generation. Only the SHA-256 hash of the secret is stored, never the secret in plain text. family_id groups all generations of the same login and parent_id chains the genealogy.

Owner service: en-monitor-security-service · **Changesets:** security-015-create-refresh-token

Field	Type	Size / values	Null	Constraint	Description
id_refresh_token	VARCHAR	10	No	PK	Token generation identifier
id_user_session	VARCHAR	10	No	FK → user_session.id_user_session	Session to which the generation belongs
family_id	VARCHAR	10	No	—	Identifier of the root of the generation family
parent_id	VARCHAR	10	Yes	FK → refresh_token.id_refresh_token	Generation that this one replaces
token_hash	VARCHAR	64	No	UNIQUE (uk_refresh_token_hash)	SHA-256 hash of the secret (never the secret in plain text)
status	ENUM	ACTIVE, ROTATED, REVOKED, EXPIRED	No	—	Lifecycle status of the generation
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
expires_at	TIMESTAMP	—	No	—	Expiration date
rotated_at	TIMESTAMP	—	Yes	—	Date on which it was replaced
revoked_at	TIMESTAMP	—	Yes	—	Revocation date
revoked_reason	ENUM	REFRESH_TOKEN_REUSE, PASSWORD_CHANGED, ADMINISTRATIVE	Yes	—	Reason for revocation
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: `ix_refresh_token_family_id` (family_id); `ix_refresh_token_user_session` (id_user_session)

security_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-security-service · **Changesets:** security-026-create-security-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	Yes	—	Event body as JSON

Field	Type	Size / values	Null	Constraint	Description
occurred_at	DATETIME	6	No	—	When the change happened (publication order)
expires_at	DATETIME	6	Yes	—	From this instant the message is no longer published (security_outbox only, for the mails with a code)
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_security_outbox_pending` (published_at, discarded_at, occurred_at)

pending_registration

Registrations waiting for the email verification. No account (person or user) exists until the code is verified; the request expires after 24 hours and an hourly job deletes it.

Owner service: en-monitor-security-service · **Changesets:** security-028-create-pending-registration

Field	Type	Size / values	Null	Constraint	Description
id_user	VARCHAR	10	No	PK	Identifier reserved for the account created when the email is verified
email	VARCHAR	255	No	UNIQUE (uq_pending_registration_email)	Email of the request (normalised to lower case)
name	VARCHAR	100	No	—	First name entered at registration
last_name	VARCHAR	100	Yes	—	Last name entered at registration (optional)
password_hash	VARCHAR	255	No	—	BCrypt hash of the chosen password
profile_image	MEDIUMTEXT	—	Yes	—	Profile picture sent at registration (base64 data URL), optional
ip_address	VARCHAR	45	Yes	—	IP address the registration came from
requested_at	DATETIME	6	No	—	When it was requested; used as the date the terms were accepted
expires_at	DATETIME	6	No	—	Expiry of the request (24 hours)

Indexes: `idx_pending_registration_expires_at` (expires_at)

appliance_type

Catalog of appliance types.

Owner service: en-monitor-device-service · **Changesets:** device-001-create-appliance-type

Field	Type	Size / values	Null	Constraint	Description
id_appliance_type	VARCHAR	10	No	PK	Type identifier
name	VARCHAR	50	No	UNIQUE (uk_appliance_type_name)	Type name

Field	Type	Size / values	Null	Constraint	Description
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

device

Records the installed measurement devices.

Owner service: en-monitor-device-service · **Changesets:** device-002-create-device, device-007-add-api-key-hash, device-008-drop-clear-api-key

Field	Type	Size / values	Null	Constraint	Description
id_device	VARCHAR	10	No	PK	Device identifier
name	VARCHAR	50	No	—	Name
appliance_type_id	VARCHAR	10	No	FK → appliance_type.id_appliance_type	Appliance type
location	VARCHAR	50	Yes	—	Location
description	VARCHAR	200	Yes	—	Description
installation_date	TIMESTAMP	—	No	—	Installation date
device_code	VARCHAR	6	No	UNIQUE (uk_device_device_code)	6-character code of the module; it is also its MQTT user name
api_key_hash	VARCHAR	64	No	—	SHA-256 hash of the integration key (the key is not stored in clear)
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

device_status_log

Records the status and connectivity of the devices.

Owner service: en-monitor-device-service · **Changesets:** device-003-create-device-status-log, device-005-status-log-last-seen-precision

Field	Type	Size / values	Null	Constraint	Description
id_device_status	VARCHAR	10	No	PK	Record identifier

Field	Type	Size / values	Null	Constraint	Description
device_id	VARCHAR	10	Yes	FK → device.id_device	Related device
status	ENUM	ONLINE, OFFLINE	No	—	Device status
signal_strength	INT	—	Yes	—	Signal strength
last_seen	TIMESTAMP	6	No	—	Last connection date
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

device_home

Relates devices to households.

Owner service: en-monitor-device-service · **Changesets:** device-004-create-device-home

Field	Type	Size / values	Null	Constraint	Description
device_id	VARCHAR	10	No	PK; FK → device.id_device	Related device
home_id	VARCHAR	10	No	PK; home (application reference, no FK)	Related household
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

device_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-device-service · **Changesets:** device-009-create-device-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	No	—	Event body as JSON
occurred_at	DATETIME	6	No	—	When the change happened (publication order)

Field	Type	Size / values	Null	Constraint	Description
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_device_outbox_pending` (published_at, discarded_at, occurred_at)

measurement

Stores the recorded electrical measurements.

Owner service: en-monitor-measurement-service · **Changesets:** measurement-001-create-measurement

Field	Type	Size / values	Null	Constraint	Description
id_measurement	VARCHAR	10	No	PK	Measurement identifier
device_id	VARCHAR	10	No	—	Device that generates the measurement
date_time	TIMESTAMP	—	No	—	Date and time of the measurement
voltage	DOUBLE	—	No	—	Measured voltage
current	DOUBLE	—	No	—	Measured current
active_power	DOUBLE	—	No	—	Active power
stored_energy	DOUBLE	—	No	—	Accumulated energy
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

consumption_level

Defines risk levels and their consumption ranges.

Owner service: en-monitor-measurement-service · **Changesets:** measurement-002-create-consumption-level

Field	Type	Size / values	Null	Constraint	Description
id_consumption_level	VARCHAR	10	No	PK	Level identifier
name	ENUM	LOW, MEDIUM, HIGH, CRITICAL	No	UNIQUE (uk_consumption_level_name)	Risk level
description	VARCHAR	200	No	—	Description

Field	Type	Size / values	Null	Constraint	Description
min_limit	DOUBLE	—	No	—	Lower limit
max_limit	DOUBLE	—	No	—	Upper limit
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

measurement_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-measurement-service · **Changesets:** measurement-004-create-measurement-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	No	—	Event body as JSON
occurred_at	DATETIME	6	No	—	When the change happened (publication order)
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_measurement_outbox_pending` (published_at, discarded_at, occurred_at); `idx_measurement_outbox_discarded` (discarded_at)

home_type

Catalog of household types.

Owner service: en-monitor-home-service · **Changesets:** home-001-create-home-type

Field	Type	Size / values	Null	Constraint	Description
id_home_type	VARCHAR	10	No	PK	Type identifier
name	VARCHAR	50	No	UNIQUE (uk_home_type_name)	Household type name

Field	Type	Size / values	Null	Constraint	Description
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

home

Records the households managed by the system.

Owner service: en-monitor-home-service · **Changesets:** home-002-create-home

Field	Type	Size / values	Null	Constraint	Description
id_home	VARCHAR	10	No	PK	Household code
name	VARCHAR	50	No	—	Household name
home_type_id	VARCHAR	10	No	FK → home_type.id_home_type	Household type
address	VARCHAR	200	No	—	Address
access_code	VARCHAR	8	No	UNIQUE (uk_home_access_code)	8-character access code (upper-case letters and digits) other users join the home with
description	VARCHAR	200	Yes	—	Description
creation_date	TIMESTAMP	—	No	—	Creation date
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

home_thresholds

Configures a household's consumption limits.

Owner service: en-monitor-home-service · **Changesets:** home-003-create-home-thresholds, home-006-add-threshold-limit-period

Field	Type	Size / values	Null	Constraint	Description
id_threshold	VARCHAR	10	No	PK	Threshold identifier
home_id	VARCHAR	10	No	UNIQUE (uk_home_thresholds_home_id); FK → home.id_home	Configured household
daily_limit	DOUBLE	—	No	—	Daily consumption limit

Field	Type	Size / values	Null	Constraint	Description
monthly_limit	DOUBLE	—	No	—	Monthly consumption limit
use_system_default	BOOLEAN	—	No	—	Uses default limits
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current
limit_period	ENUM	DAILY, MONTHLY	No	DEFAULT 'DAILY'	Limit set by the owner (DAILY or MONTHLY)

Indexes: None

user_home

Relates users to households and defines their role.

Owner service: en-monitor-home-service · **Changesets:** home-004-create-user-home

Field	Type	Size / values	Null	Constraint	Description
user_id	VARCHAR	10	No	PK; user (application reference, no FK)	Related user
home_id	VARCHAR	10	No	PK; FK → home.id_home	Related household
role	ENUM	OWNER, MEMBER	No	—	User's role in the household
favorite	BOOLEAN	—	No	—	Indicates whether it is the favorite household
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

home_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-home-service · **Changesets:** home-007-create-home-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	No	—	Event body as JSON
occurred_at	DATETIME	6	No	—	When the change happened (publication order)

Field	Type	Size / values	Null	Constraint	Description
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_home_outbox_pending` (published_at, discarded_at, occurred_at)

notification

Record of the delivery of a message to the user. Alert and Recommendation are events; Notification records whether the message was delivered, through which channel and its read status.

Owner service: en-monitor-notification-service · **Changesets:** notification-001-create-notification, notification-004-add-in-app-admin-notifications, notification-006-add-activity-sources, notification-007-add-message-params

Field	Type	Size / values	Null	Constraint	Description
id_notification	VARCHAR	10	No	PK	Notification identifier
user_id	VARCHAR	10	No	user (application reference, no FK)	Recipient user
home_id	VARCHAR	10	Yes	home (application reference, no FK)	Related home
channel	ENUM	EMAIL, PUSH, IN_APP	No	—	Channel: EMAIL, PUSH or IN_APP (inbox of the web and the app)
source_type	ENUM	ALERT, RECOMMENDATION, SECURITY, ADMINISTRATION, ACCOUNT, HOME, DEVICE	No	—	Source: ALERT, RECOMMENDATION, SECURITY, ADMINISTRATION, ACCOUNT, HOME or DEVICE
source_id	VARCHAR	10	Yes	(source_type) (application reference, no FK)	Origin identifier
message_key	VARCHAR	100	No	—	Message key
delivery_status	ENUM	PENDING, SENT, FAILED	No	—	Delivery status
sent_at	TIMESTAMP	—	Yes	—	Send date
read_at	TIMESTAMP	—	Yes	—	Read date
failure_reason	VARCHAR	200	Yes	—	Reason for delivery failure
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Field	Type	Size / values	Null	Constraint	Description
message_params	TEXT	—	Yes	—	Text parameters as JSON (names of the home, the device or the people at the time of the event), so the client translates the text

Indexes: `idx_notification_user_channel_created` (user_id, channel, created_at); `idx_notification_source` (source_type, source_id)

notification_preference

Notification channel preferences of each user (email and push).

Owner service: en-monitor-notification-service · **Changesets:** notification-002-create-notification-preference

Field	Type	Size / values	Null	Constraint	Description
user_id	VARCHAR	10	No	PK; user (application reference, no FK)	User who owns the preferences (primary key)
email_enabled	BOOLEAN	—	No	DEFAULT TRUE	Delivery of alerts and recommendations by email
push_enabled	BOOLEAN	—	No	DEFAULT TRUE	Delivery through browser notification
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: None

push_subscription

Push notification subscriptions of the user's browsers.

Owner service: en-monitor-notification-service · **Changesets:** notification-003-create-push-subscription

Field	Type	Size / values	Null	Constraint	Description
id_push_subscription	VARCHAR	10	No	PK	Subscription identifier
user_id	VARCHAR	10	No	user (application reference, no FK)	Subscribed user
endpoint	VARCHAR	500	No	UNIQUE (uk_push_subscription_endpoint)	Address of the browser push service (unique)
p256dh	VARCHAR	100	No	—	Public encryption key of the subscription
auth	VARCHAR	50	No	—	Authentication secret of the subscription
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: `idx_push_subscription_user_id` (user_id)

push_device

Mobile app installations that receive push notifications through the Expo relay (one row per installation). It is not another kind of PushSubscription: a relay token carries no endpoint or encryption keys.

Owner service: en-monitor-notification-service · **Changesets:** notification-005-create-push-device

Field	Type	Size / values	Null	Constraint	Description
id_push_device	VARCHAR	10	No	PK	Installation identifier
user_id	VARCHAR	10	No	user (application reference, no FK)	User who owns the installation (indexed)
token	VARCHAR	255	No	UNIQUE (uk_push_device_token)	Push token of the installation; registering it again changes its owner
platform	VARCHAR	10	No	—	Platform of the installation (the published app is Android: ANDROID)
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: `idx_push_device_user_id` (user_id)

alert

Alerts of a home: reading above the power threshold (THRESHOLD, level HIGH or CRITICAL), daily or monthly limit reached (LIMIT), device offline (CONNECTIVITY) and first reading of a newly linked device (DEVICE).

Owner service: en-monitor-alert-service · **Changesets:** alert-001-create-alert, alert-002-add-device-alert-type, alert-003-add-limit-alert-type

Field	Type	Size / values	Null	Constraint	Description
id_alert	VARCHAR	10	No	PK	Alert identifier
home_id	VARCHAR	10	No	home (application reference, no FK)	Household it applies to
device_id	VARCHAR	10	Yes	device (application reference, no FK)	Related device
type	ENUM	THRESHOLD, CONNECTIVITY, DEVICE, LIMIT	No	—	Alert type: THRESHOLD, CONNECTIVITY, DEVICE or LIMIT
message_key	VARCHAR	100	No	—	Key of the text the client translates (alert.threshold.high, alert.threshold.critical, alert.limit.daily, alert.limit.monthly, alert.connectivity.offline, alert.device.linked)
date_time	TIMESTAMP	—	No	—	Date and time
alert_status	ENUM	PENDING, RESOLVED	No	—	PENDING while active; RESOLVED once resolved or marked as read

Field	Type	Size / values	Null	Constraint	Description
consumption_level_id	VARCHAR	10	Yes	consumption_level (application reference, no FK)	Associated consumption level
measurement_id	VARCHAR	10	Yes	measurement (application reference, no FK)	Associated measurement
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: `idx_alert_home_id` (home_id)

alert_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-alert-service · **Changesets:** alert-004-create-alert-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	No	—	Event body as JSON
occurred_at	DATETIME	6	No	—	When the change happened (publication order)
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_alert_outbox_pending` (published_at, discarded_at, occurred_at)

recommendation

Saving recommendations the recommendation service generates every hour from the hourly energy of the home; device_id empty when it applies to the whole home.

Owner service: en-monitor-recommendation-service · **Changesets:** recommendation-001-create-recommendation

Field	Type	Size / values	Null	Constraint	Description
id_recommendation	VARCHAR	10	No	PK	Recommendation identifier

Field	Type	Size / values	Null	Constraint	Description
home_id	VARCHAR	10	No	home (application reference, no FK)	Household it applies to
device_id	VARCHAR	10	Yes	device (application reference, no FK)	Related device
type	ENUM	SAVING, PEAK_HOURS, HIGH_CONSUMPTION, HISTORICAL_COMPARISON, THRESHOLD	No	—	Rule that produced it: PEAK_HOURS, SAVING, HISTORICAL_COMPARISON, THRESHOLD or HIGH_CONSUMPTION
message_key	VARCHAR	100	No	—	Text key (recommendation.peakHours, .standby, .aboveAverage, .limitProjection, .deviceIncrease)
date_time	TIMESTAMP	—	No	—	Date and time
status	ENUM	READ, UNREAD	No	—	UNREAD or READ
created_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP	Creation timestamp of the row (audit)
updated_at	TIMESTAMP	—	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp of the last change (audit)
deleted_at	TIMESTAMP	—	Yes	—	Soft-delete timestamp; empty while the row is current

Indexes: `idx_recommendation_home_id` (home_id)

recommendation_outbox

Transactional outbox of the service: every event the service publishes to RabbitMQ is written here in the same transaction as the change, and a relay publishes it every second, oldest first.

Owner service: en-monitor-recommendation-service · **Changesets:** recommendation-002-create-recommendation-outbox

Field	Type	Size / values	Null	Constraint	Description
id_message	VARCHAR	10	No	PK	Message identifier; travels as the RabbitMQ message id so consumers can drop duplicates
message_type	VARCHAR	60	No	—	Message type (routing key), for example `home.deleted`
payload	TEXT	—	No	—	Event body as JSON
occurred_at	DATETIME	6	No	—	When the change happened (publication order)
published_at	DATETIME	6	Yes	—	When the broker confirmed the message; empty while pending
discarded_at	DATETIME	6	Yes	—	When the message was discarded without being published (expired, or written by the tests)
attempts	INT	—	No	DEFAULT 0	Publication attempts made
last_error	VARCHAR	255	Yes	—	Last publication error (for example, no queue took the message)

Indexes: `idx_recommendation_outbox_pending` (published_at, discarded_at, occurred_at)

5. Seed data

Table	Changeset	Content
password_policy	security-019	Default policy `pol0000001`: 8 to 64 characters, upper case, digits and symbols required, 90-day expiry (only when the table is empty).
system_role, permission, system_role_permission	security-025	Global role `ADMIN` (`rol0000001`) with the permissions `USER.READ` (`prm0000001`), `USER.MANAGE` (`prm0000002`) and `ROLE.MANAGE` (`prm0000003`). No account gets the role: the first administrator is granted with `make-admin.sh` of the energy-monitor-infra repository.
appliance_type	device-006	refrigerator, air_conditioner, lighting, washing_machine, television, microwave, computer, water_heater and other (`appl000001` to `appl000009`).
consumption_level	measurement-003	LOW [0, 500) W, MEDIUM [500, 1500) W, HIGH [1500, 3000) W and CRITICAL [3000, 1,000,000) W of active power.
home_type	home-005	house, apartment, studio, country_house, cabin and other.

The default limits of a home (6 kWh per day and 180 kWh per month) are not seed data: they are configuration of the home service (`daily-limit` and `monthly-limit` in its `application.yaml`).